



CST413-QP-1 - question paper

Machine Learning (APJ Abdul Kalam Technological University)



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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree 7th semester (S,FE) Exam April 2025 (2019 Scheme)

Course Code: CST413

Course Name: MACHINE LEARNING

Max. Marks: 100

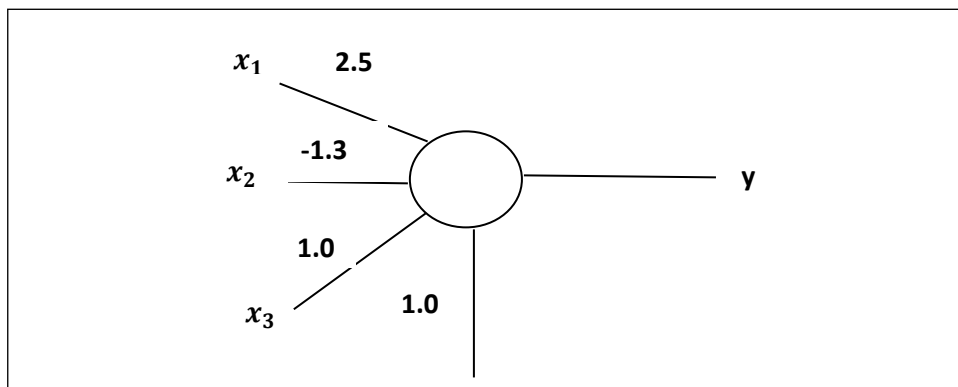
Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- 1 Identify the suitable learning method in each case and explain it (3)
 - a) Music recommender system can suggest songs to users based on their listening pattern.
 - b) Disease Prediction
- 2 State noise in the data? How it affects the results? (3)
- 3 Compute ML estimate for the parameter p in the binomial distribution whose probability function is $f(x) = \binom{n}{x} p^x (1-p)^{(n-x)}$, where $x = 0, 1, 2, \dots, n$ (3)
- 4 What are the different metrics used in deciding the splitting attribute? (3)
- 5 The figure shows a neuron with activation function $f(u) = \frac{0.6}{1 + \exp(10.2u)}$ where u indicates total synaptic input. Explain the figure with respect to neural network algorithm (3)



- 6 Suppose that you have a linear support vector machine(SVM) binary classifier. (3)

Consider a point that is currently classified correctly, and is far away from the decision boundary. If you remove the point from the training set, and re-train the

classifier, will the decision boundary change or stay the same? Justify your answer

- 7 Compare K-means algorithm and hierarchical clustering. (3)
- 8 Principal Component Analysis (PCA) is used for Dimensionality Reduction. Justify your answer. (3)
- 9 Describe boosting. Relate boosting and ensemble learning? (3)
- 10 Describe Precision and Recall (3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- 11 a) Suppose that X is a discrete random variable with the following probability mass function: where $0 \leq \theta \leq 1$ is a parameter. The following 10 independent observations were taken from such a distribution: (3,0,2,1,3,2,1,0,2,1). What is the maximum likelihood estimate of θ (7)

X	0	1	2	3
$P(X)$	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

- b) Let $X_1, \dots, X_n$ be an i.i.d. sample from a Poisson distribution with parameter λ . Find the MLE of the parameter λ . (7)

OR

- 12 a) Let X be a single observation from a Binom(n, p), where p is an unknown parameter. (In this case, we will consider n known.) Find the maximum likelihood estimator (MLE) of p (7)
- b) Suppose you roll a 6-sided die 40 times and observe eight rolls of a 6. What is the maximum likelihood estimate of the probability of observing a 6? (3)
- c) Using the same observed data, suppose you now plan to perform a second experiment with the same die, and will roll the die 5 more times. What is the maximum likelihood estimate of the probability that you will observe no 6's in this next experiment? (4)

Module II

- 13 a) Discuss the decision tree algorithm with an example (8)
- b) Illustrate the use of decision tree as classification (6)

OR

- 14 a) Given the following table of data (9)

T_id	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	Y	No
2	No	Married	Y	No
3	No	Single	N	No
4	Yes	Married	Y	No
5	No	Divorced	N	Yes
6	No	Married	N	No
7	Yes	Divorced	Y	No
8	No	Married	N	No
9	No	Married	N	No
10	No	Single	N	Yes

Find out the probability of the attribute values X:(Refund = No, Status = single, Taxable income = No) to belong to the class = yes and class = no

- b) The sales of a company (in million dollars) for each year are shown in the table below (5)

X (Years)	2005	2006	2007	2008	2009
Y(sales)	12	19	29	37	45

- a) Find the least square regression line
 b) Use the least square regression line as a model to estimate the sales of the company in 2012.

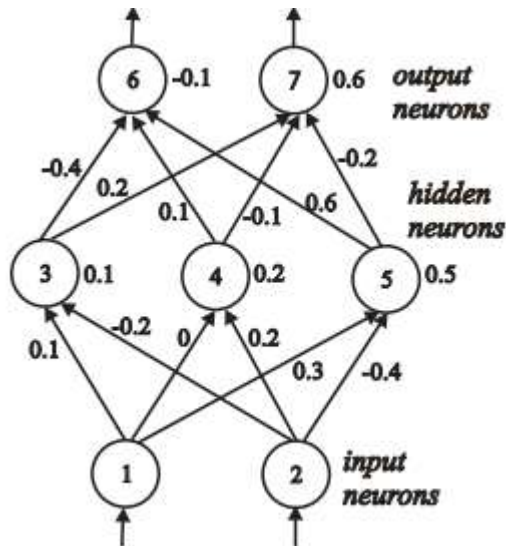
Module III

- 15 a) List out the parameters involved in back propagation network and explain the algorithm (8)
 b) Discuss the issues of convergence, local minima and generalization, overfitting and stopping criteria (6)

OR

- 16 a) Why SVM is not appropriate for non-linear data? Choose an appropriate technique to make SVM suitable for non-linear data with appropriate example. (4)
 b) Given the following neural network with initialized weights as in the picture, (10)

explain the network architecture knowing that we are trying to distinguish between nails and screws and an example of training tuples is as follows:
 $T1\{0.6, 0.1, \text{nail}\}$, $T2\{0.2, 0.3, \text{screw}\}$.



Let the learning rate η be 0.1 and the weights be as indicated in the figure above. Do the forward propagation of the signals in the network using T1 as input, then perform the back propagation of the error. Show the changes of the weights

Module IV

- 17 a) Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters: $A1 = (2, 10)$, $A2 = (2, 5)$, $A3 = (8, 4)$, $A4 = (5, 8)$, $A5 = (2, 10)$, $A6 = (6, 4)$, $A7 = (1, 2)$, $A8 = (4, 9)$. (8)
- b) Illustrate the strength and weakness of k-means algorithm. (6)

OR

- 18 a) Explain the procedure for the computation of the principal components of the data. (6)
- b) What is the significance of kernels in SVM? What are the different types of kernels? (8)

Module V

- 19 a) Given the Following data about True Label and Predicted Labels (6)

Data point	True Labels	Predicted Labels
0	1	1
1	1	-1
2	1	-1
3	1	-1
4	-1	1
5	-1	1
6	-1	1
7	-1	-1
8	-1	1
9	-1	-1
10	-1	-1
11	1	1
12	-1	1
13	-1	-1

Compute (a) Precision (b) Recall (c)F1 score

- b) Explain the significance of ROC curve and AUC curve (8)

OR

- 20 a) Given the following distance matrix, construct dendrogram using single linkage, hierarchical clustering algorithm (6)

Item	A	B	C	D	E
A	0	9	3	6	11
B	9	0	7	5	10
C	3	7	0	9	2
D	6	5	9	0	8
E	11	10	2	8	0

- b) Define the following (8)

- i. k-fold Cross-validation
- ii. Holdout Method
- iii. Leave-one-out Method
- iv. Bootstrap
